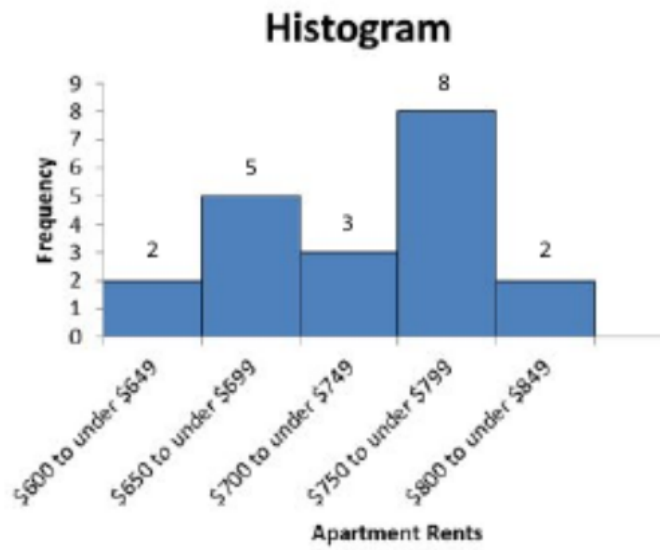
Worksheet 3 Solution

Fred Azizi

2023-02-21

Remaining from From Worksheet 2:

2. The following histogram shows the distribution of the monthly rental for a random sample of one-bedroom apartments in York, Pennsylvania.



a. What is the total number of apartments in this sample, and what is the percentage of monthly rents that are $750 and above? If we rank the observations from low to high, what can you say about the range of 8th ranked observation?

<u>Answer:</u>

Total number of apartment in this sample is 2+5+3+8+2= 20.

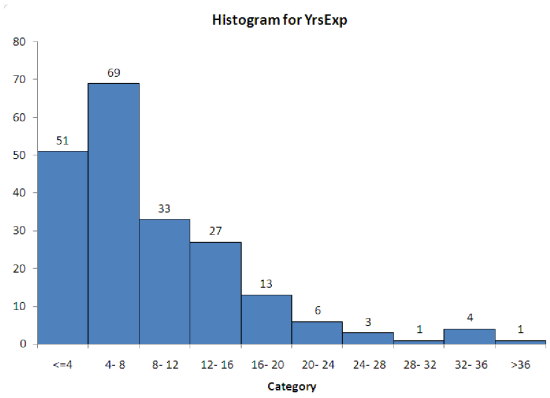
We need to look at the last two rectangles in the histogram. They represent a rent of 750 or greater. The total frequency for these two classes would be 8+2=10. In this case, percent frequency would be $\frac{10}{20} \times 100 = 50\%$.

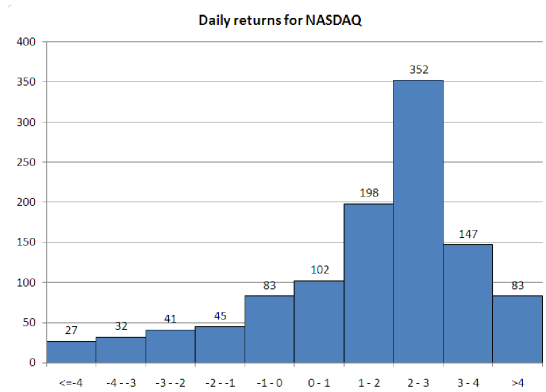
The 8th observation will be between 700 and 749 as the cumulative total frequency up to class of 700 to 749 is 7.

b. Suppose the rent for one of the apartments in the sample is $680. How many apartments rent in the sample are going to be lower than or equal to this rent number?

Answer:

c. Identify the shape of below histograms [1]





Answer:

Plot on the left (histogram for years of experience) has a long tail on the right side. Hence, positively skewed.

Plot on the right (histogram for Daily returns for NASDAQs) has a long tail on the left side. Hence, negativity skewed.

3. The U.S. National Debt over the span of a decade from 1991 to 2001 is given in the following table:

Year	Debt (in T)
1991	7.3
1992	7.9
1993	8.3
1994	8.6
1995	8.9
1996	9.1
1997	9.2
1998	9.3
1999	9.2
2000	9.0
2001	8.9

a. Is this an example of time series data or cross sectional data?

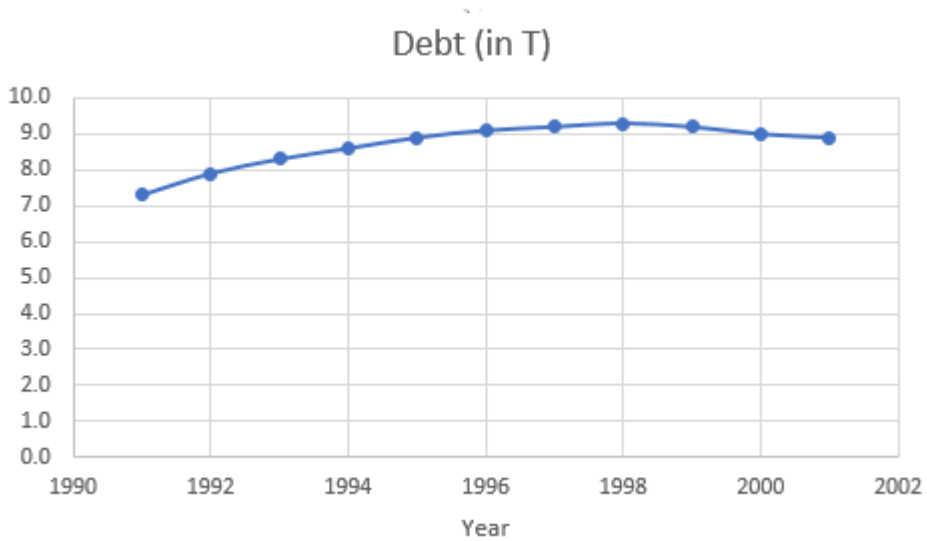
Answer:

This is an example of time series data.

b. Make an appropriate plot for this data.

[1] Graphs are from http://citadel.sjfc.edu/faculty/kgreen/MSTI130/MSTI130Text/Text_Fall_2014su28.html

Answer:



c. What can you conclude from this data?

Answer:

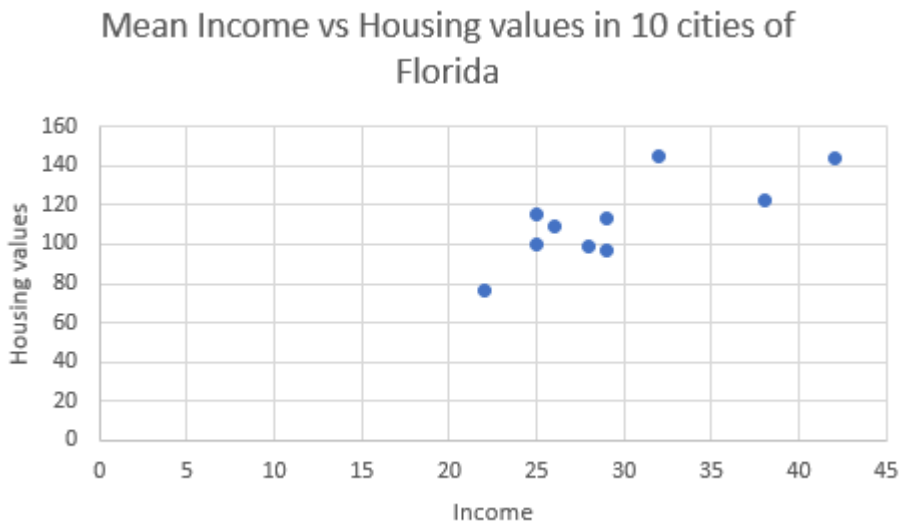
We can see a slight increase in debt over the first half of the decade and after that it stayed almost the same over the second half. Note that **we can't use the graph of a time series over a period of time for making conclusions about future**. This decade shows no strong trend, but in the following years the U.S. debt increased dramatically due to wars.

4. The following data has mean income and housing for 10 cities in Florida. Values are in dollars ($) and rounded to the nearest thousand.

City	Income ($\boldsymbol{x}$)	Housing (y)
A	26	109
B	29	97
C	25	115
D	28	99
E	38	122
F	32	145
G	25	100
H	22	76
I	29	113
J	42	144

a. Draw an appropriate diagram representing the relationship between Income (x) and Housing (y).

Answer:



b. Suppose the rent for one of the apartments in the sample is \$680. How many apartments rent in the sample are going to be lower than or equal to this rent number?

Answer:

This could be anywhere from 2 to 6. \$680 is inside the second class, so it will be bigger than the 2 observations in the first class. But we don't know its relative position inside the second class. If it is the smallest number, then only the 2 observation in the first class will be lower than 680, but if it is the biggest number in the second class, 4+2 observartions could be smaller than 680.

c. How would you describe the relationship between the income (x) and housing (y)?

Answer:

Points in the scatterplot are oriented toward the top right of the plot. Therefore, income and housing have a positive relationship.

5. A random sample of 10 students were asked how far they live from UMBC. These were their answers, rounded to the nearest half-mile:

$$1, 2.5, 1.5, 2, 0, 5, 3, 3.5, 1, 0.5$$

Compute the mean, median and mode for this.

Answer:

Mean $= \bar{x} = \frac{\sum x}{n} = \frac{20}{10} = 2$.

Median: Order the data points from low to high: 0, 0.5, 1, 1, 1.5, 2, 2.5, 3, 3.5, 5. $\Rightarrow$ Position of median $= (n+1)\frac{50}{100} = 11 \times 0.5 = 5.5$ th position $\Rightarrow$ Median $= (1.5+2)/2 = 1.75$

Mode = 1 (occurs twice).